YOLO-ORE: A Deep Learning-Aided Object Recognition Approach for Radar Systems

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***Abstract*—To enable intelligent vehicles and transportation systems, the vehicles and relevant systems need to have the ability to sense environment and recognize objects. In order to benefit from the robustness of radar for sensing, knowing how to use the radar system for effective object recognition is critical. Observing this, we in this paper propose a novel deep learning-aided object recognition system for radar systems by combining the You only look once (YOLO) system with a proposed object recheck system. Our proposed system is able to benefit from conventional YOLO and also mitigate the overlap errors and misclassification errors induced by using YOLO. We conduct extensive real-world experiments in realistic scenarios to evaluate our proposed object recognition system. Results validate that our system can provide good performance in complicated real-world scenarios. The results also show that our proposed object recognition system can outperform the state-of-the-art learning-based object recognition systems.**

***Index Terms*—Automotive radars, deep learning, object detection, object recognition, FMCW.**

NOMENCLATURE

Frequently Used Notations

$\mathbf{M}^{\text{RDA}}$	The Range-Doppler-Angle (RDA) map.
S_{max}	The maximum signal strength of all RA maps.
N_{f}	The total number of measurement frames.
S_{UB}	The lower bound of the signal strength.
$\mathbf{M}^{\text{ERA}}$	The enhanced RA (ERA) map.
x	The centering x-coordinate of the bndbox.
l	The length of the bndbox.
C	The class of the object in the bndbox.
$T_{\text{C,YOLO}}$	The confidence threshold.
IOU	The intersection over union.
A_{ij}^{union}	The size of union of bndboxes of two objects i and j
IOB	The intersection over bndbox.
$T_{\text{IOB,NMSDC}}$	The given IOB threshold for NMSDC.
y_{CB}	The y-direction centering location of the cluster bndbox on the Cartesian coordinate system.
w_{CB}	The width of the cluster bndbox.
$Y_{\text{CB}}^{\text{max}}$	The y-axis value of the top-most radar point of the cluster.
$Y_{\text{CB}}^{\text{min}}$	The y-axis value of the bottom-most radar point of the cluster.
L_{CB}	The length of radar points in the object of interest.
S_{CB}	The shape-volume of radar points in the object of interest.
$Y_{\text{CB}}^{\text{SD}}$	The standard deviation of the y-axis value for all points in the object.
SNR_i	The estimated SNR given by the radar system for the radar point i of the object.
$V_{\text{CB}}^{\text{Diff}}$	The maximum difference velocity of the object.
$V_{\text{CB}}^{\text{max}}$	The maximum velocity of the radar points of the object.
$\mathbf{M}^{\text{RA}}$	The Range-Angle (RA) map.
S_{min}	The minimum signal strength of all RA maps.
$\mathbf{M}^{\text{NRA}}$	The normalized RA (NRA) map.
S_{LB}	The upper bound of the signal strength.
x_e	The parameter that controls the highlighting of the targets.
y	The centering y-coordinate of the bndbox.
w	The width of the bndbox.
Class	The corresponding confidence level.
$T_{\text{IOU,YOLO}}$	The intersection over union (IOU) threshold.
A_{ij}^{inter}	The size of intersection of two objects i and j
$T_{\text{LC,ORE}}$	The low-confidence threshold for ORE system.
K_{OVL}	The number of objects overlapping with one another.
x_{CB}	The x-direction centering location of the cluster bndbox on the Cartesian coordinate system.
l_{CB}	The length of the cluster bndbox.
$X_{\text{CB}}^{\text{max}}$	The x-axis value of the rightmost radar point of the cluster.
$X_{\text{CB}}^{\text{min}}$	The x-axis value of the leftmost radar point of the cluster.
n_{CB}	The number of radar points in the object of interest.
W_{CB}	The width of radar points in the object of interest.
$X_{\text{CB}}^{\text{SD}}$	The standard deviation of the x-axis value for all points in the object.
D_{CB}	The intensity feature of the object.
$\overline{V}_{\text{CB}}$	The average velocity of the object.

Manuscript received 19 May 2022; revised 15 October 2022; accepted 1 December 2022. Date of publication 26 December 2022; date of current version 18 May 2023. This work was supported in part by the National Science and Technology Council of Taiwan under Grants 111-2218-E-A49-024, 111-2221-E-A49-070, 111-3114-E-A49-001, and 110-2221-E-A49-025-MY2, and in part by Qualcomm through a Taiwan University Research Collaboration Project. Part of the dataset collected and used in this paper is online assailable for non-commercial use in https://ieee-dataport.org/documents/yoloore. Redistribution prohibited. The review of this article was coordinated by Prof. Gabriella Olmo.
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Digital Object Identifier 10.1109/TVT.2022.3232135

$V_{\mathrm{CB}}^{\mathrm{SD}}$ The velocity standard deviation of the object.
$V_{\mathrm{CB}}^{\max}$ The minimum velocity of the radar points of the object.

I. Introduction

IN RECENT years, thanks to the evolution of different technologies, e.g., the internet of vehicles (IoVs) [1], computing technology [2], sensing technology [3], and machine learning [4], both the academia and industry are envisioning that the vehicles and transportation systems can be made intelligent [5], [6]. Among all the critical technologies for intelligent vehicles and transportation systems, how to give vehicles and transportation systems the ability to sense the environment and recognize events and objects is one of them. Therefore, how to realize object recognition using sensing systems has recently been intensively discussed [6]. According to the type of sensors, object recognition techniques can be mostly distinguished into three different categories: camera-based, LiDAR-based, and radar-based recognition techniques [7], [8]. The camera-based object recognition is the most commonly used approach for performing the object recognition thanks to the rapidly increasing number of cameras and the development of computer vision [9], [10], [11]. However, since it is a vision-based approach that relies on the camera image to capture the features for object recognition, such an approach can be vulnerable to bad weather conditions, e.g., heavy rain and thick fog. On the other hand, LiDAR-based object recognition uses LiDAR point cloud to extract features for object recognition [12], [13], [14]. Although the LiDAR-based approaches can provide high accuracy via using dense light pulses, it still suffers from high equipment cost and computing complexity. Furthermore, it also could be vulnerable to bad weather conditions due to the use of visible light for sensing.

Using radar sensors to estimate targets has been investigated for many years. The evident results showed that radar sensing can provide some necessary information about the targets even in harsh environments [15], [16]. Besides, because radar systems have been widely investigated, the cost of installing a high-performance radar is relatively low as compared to LiDAR. Due to the high reliability in bad weather and the low-cost characteristic of radar sensing, the radar-based object recognition could be a promising candidate for realizing the necessity of object recognition.[1] Thus, it has recently started to draw attention [17], [18].

A. Literature Review

In this paper, we focus on investigating the radar-based object recognition. Because using learning-based solutions for object recognition has been a great success with camera [19], investigations that use learning-based solutions to conduct the object recognition with radar have recently been popular [20], [21], [22], [23], [24], [25], [26]. We note that an object recognition task in general includes the object detection and the object classification. In [20], the authors proposed the Deepreflecs that applies the clustered point cloud to improve the object classification accuracy without improving the object detection and clustering performance. In [21], considering moving objects, a point-based neural network (NN) was performed directly on radar points so that the object detection and classification performance can be improved. Also with the radar points, [22] proposed using support vector machine (SVM) to conduct object classification for static and moving objects. In contrast to [21] and [22], authors in [23] considered using a convolutional NN (CNN) with radar grids to conduct object detection and classification for static objects. To conduct the object recognition for both static and moving objects, in [24], an object recognition approach was proposed by combining approaches in [21] and [23].

Because the radar point cloud is a compressed representation for the radar Range-Angle (RA) map, the above approaches that use radar point cloud to conduct object recognition basically could be limited by the information loss due to the conversion from radar RA map to point cloud. To remedy this, the authors in [25] and [26] proposed to apply machine learning solutions directly to radar maps so that the overall radar sensing information can be fully exploited. We note that although the aforementioned works have shown that it is feasible to use learning-based solutions for radar systems to conduct the object recognition, they might suffer from the drawback that the direct use of the features extracted during the radar processing might not provide the best performance and the conversion of radar data to features for object recognition requires additional overhead. In addition, as they use different radar processing modules to extract features for different recognition purposes, they cannot provide complete information about the objects, e.g., locations, shapes, categories, and sizes of objects, by using in a single object recognition task, leading to additional troublesome of the information and system integration.

Considering the disadvantages of the above approaches, one-stage NN models for object recognition were discussed in [19]. In addition, [27] and [28] validated that instead of using multiple radar signal processing modules and learning solutions for completing the object recognition task, using a NN to complete the task in one-shot is feasible and can even provide better performance. Recently, it has been shown that the you only look once (YOLO) system, which is a well-known one-stage NN model, can provide the comprehensive information about objects (locations, shapes, categories, and sizes of objects), and is very powerful for object recognition using camera images [29]. Thus, studies on how to apply the YOLO system to radar object recognition have recently been widely discussed [30], [31], [32], [33], [34], [35], [36], [37], [38].

To be specific, in [30], a basic YOLO was performed on the Range-Doppler (RD) map to recognize the pedestrian, cyclist, and car. In [31], the authors considered using a low-complexity YOLO and showed that the object recognition can be conducted by using a light YOLO on RD maps. In [32], the YOLO system was applied to the RA map and showed that using the YOLO

[1] We note that, however, the radar-based object recognition is not aiming to completely replace the other two types of object recognition. Instead, it serves as a very good compensation for them when encountering different situations. The integration and fusion of different results from different types of sensors indeed is a promising solution to enable an intelligent system that can deal with very complicated environments.

with RA maps is better than using the YOLO with RD maps. In [33], to recognize the pedestrian, car, and trailer, a YOLO system was performed on Range-Range (RR) maps obtained by transforming the radar RA map results to the corresponding results with Cartesian coordinate system. Different from the above works that use the complete radar RA or RD map, [34] and [35] conducted the object recognition using YOLO with processed radar point clouds. To further enhance [34] and [35], a more advanced version of YOLO, called ComplexYOLO, was proposed to perform object recognition using three-dimensional (3D) radar point clouds [36]. To obtain the benefits from both using radar maps and using radar point clouds, [37] proposed a structure that combines YOLO with the SVM, called YOLO-SVM, so that both the RR map and the radar point cloud can be used for object recognition. In [38], a combination of YOLO and a learning-based tracking system was proposed for conducting both the object recognition and tracking. The above investigations showed that using YOLO system for radar object recognition is promising.

B. Contributions

Based on the above literature review and our own experimental experiences, we observe that the study of using YOLO for object recognition is yet to be complete. Specifically, although [37] has shown that the best performance of radar object recognition can be obtained via using YOLO with both the RR map and radar point cloud, the discussion in [37] was restricted to the object recognition considering only pedestrians and vehicles in simple scenarios. In addition, by our experiments, we found that the objects detected by [37] can be overlapped with one another at the same position. However, since different objects commonly cannot physically overlap with one another at the same position, this leads to overlapping errors. Furthermore, none of the previous papers that study using YOLO for object recognition evaluate and validate the YOLO-based object recognition approach using extensive real-world experiment data. By these observations, we thus in this paper aim to develop a YOLO-based radar object recognition approach that can deal with the above drawbacks and validate its performance using extensive real-world experiment data.

We in this paper consider investigating the YOLO-based object recognition approach for radar systems so that the objects can be detected with the corresponding locations, shapes, categories, and sizes. In this paper, we propose a deep learning-aided object recognition system, named YOLO-ORE (YOLO-Object REcheck), that consists of three main functions: radar image generation, YOLO-based object recognition, and object recheck. In radar image generation, we propose a radar image generation method, called target highlight technique (THT), that can convert the RA map generated by the radar to the radar image (RR map) with possible targets being highlighted. Then, the radar image is fed into a YOLO model to generate the basic object recognition result of targets. Since the result provided by YOLO can have overlapping and misclassification errors caused by having detected objects unreasonably overlapped with one another and by incorrectly detecting the targets, respectively, the object recognition result from YOLO is further fed into an object recheck system to eliminate overlapping and misclassification errors. By using a standard radar module, we conduct extensive real-world experiments to validate our system in both simple and complex scenarios (in total more than 14,500 radar images with more than 30,000 objects). Results show that our proposed YOLO-ORE system for object recognition can provide good performance and outperform the state-of-the-art YOLO-based object recognition system in all scenarios. To the best of our knowledge, this paper is the first paper to evaluate and validate the YOLO-based recognition approach with such an extensive dataset whose data is from both simple and complex scenarios.

In summary, our main contributions of this paper are two-fold:

- We propose a novel YOLO-based object recognition approach that can effectively detect the targets with the corresponding locations, shapes, categories, and sizes with radar systems.
- We conduct extensive real-world experiments to evaluate our proposed approach in both simple and complex scenarios. Results show that our proposed approach can provide good performance and outperform the state-of-the-art approach. We stress that this is the first paper to provide the evaluation with such an extensive real-world dataset.

C. Organization

The remainder of this paper is organized as follows. In Section II, the frequency modulated continuous waveform (FMCW) radar signal processing is briefly reviewed. In Section III, our proposed YOLO-ORE system is presented with details. The descriptions for the real-world experiments and the evaluation results are provided in Section IV. Finally, we conclude this paper in Section V. Some additional comparison results that are less critical are relegated to Appendix.

II. FMCW Radar Signal Model

Our proposed deep learning-aided object recognition approach uses the information from the radar system, namely, the radar RA map and point cloud, to conduct the object recognition. Therefore, in this section, we briefly introduce the radar signal processing for obtaining the RA map and point cloud based on the FMCW radar system. Note that although we consider the FMCW radar here due to that we use it to conduct the experiments in Section IV, our proposed approach can be used with other types of radar systems as long as the RA map and the point cloud can be obtained by the systems.

A. Radar Signal Processing

The FMCW radar periodically transmits the frequency-modulated signal whose carrier frequency increases linearly over a period of time, where a complete transmission of a period is called a chirp. Considering a single chirp, the instantaneous frequency of the transmitted chirp is given as:

$$f_{\text{tx}}(t) = f_c + \frac{B}{2T}t, \quad 0 \leq t \leq T, \tag{1}$$

where f_c is the starting carrier frequency of a chirp, $\frac{B}{2^{\sim}T}$ is the frequency slope of a chirp, B is the maximum sweep bandwidth, and T is the chirp duration. With (1), the transmitted signal of a chirp can be expressed as:

$$s(t) = e^{j2\pi f_{\text{tx}}(t)t} = e^{j2\pi (f_c + \frac{B}{2^{\sim}T} t)t}, \quad 0 \le t \le T. \tag{2}$$

Then, assuming that there exist H targets, the corresponding received radar signal can be expressed as:

$$r(t) = \sum_{h=1}^{H} b_h s(t - \tau_h(t)) e^{j\phi_h} + w(t), \quad 0 \le t \le T, \tag{3}$$

where b_h, ϕ_h, and τ_h are attenuation factor, random phase, and round-trip time associated with target h, respectively, and $w(t)$ is the additive white Gaussian noise (AWGN). To extract the distance and relative velocity information of the targets embedded in the received signal, the received signal would be mixed with its transmitted signal at the radar receiver, leading to:

$$\begin{aligned} g(t) &= r(t) \cdot s^*(t) \\ &= \sum_{h=1}^{H} b_h e^{-j2\pi (f_c + \frac{B}{T}(t - \tau_h(t)))\tau_h(t)} e^{j\phi_h} + \breve{w}(t), \quad 0 \le t \le T, \end{aligned} \tag{4}$$

where $(\cdot)^*$ is the complex conjugate operation and $\breve{w}(t)$ is the noise after the mixing. We denote the distance between target h and radar and the velocity of the target h relative to the radar, namely, the range and Doppler velocity, as R_h and v_h, respectively. Then, τ_h can be approximately given as:

$$\tau_h = \tau_h(t) = \frac{2(R_h + v_h t)}{c}, \quad 0 \le t \le T, \tag{5}$$

where c is the speed of light. To process $g(t)$, we conduct sampling with frequency $f_s = \frac{1}{T_s}$. Thus, the sampled signal is given as:

$$\begin{aligned} g[n] = g(nT_s) = \sum_{h=1}^{H} b_h e^{-} j2\pi (f_c \\ + \frac{B}{T}(nT_s - \tau_h))\tau_h e^{j\phi_h} + \breve{w}(nT_s), \quad 0 \le n \le N-1, \end{aligned} \tag{6}$$

where $N = \lfloor \frac{T}{T_s} \rfloor$ is the total number of samples in a chirp.

With a single chirp, we only can detect the range, while the Doppler velocity cannot be effectively detected as $v_h T$ is too small to be observed within a chirp duration. Therefore, the complete radar system would transmit multiple chirps within a detection period. Suppose that there are M chirps transmitted consecutively. By repeating the above process, we can obtain NM samples in a detection period. By letting the element of row k and column l in a $N \times M$ matrix $\mathbf{F}$ be the sample k in chirp l and letting $\Delta f = \frac{BT_s}{T}$, the element of row k and column l in $\mathbf{F}$ can be expressed as [40]:

$$\begin{aligned} (\mathbf{F})_{k,l} = \sum_{h=1}^{H} b_h e^{-j2\pi T f_{D,h} l} e^{-j2\pi \tau_h k \Delta f} e^{j\hat{\phi_h}} + (\mathbf{W})_{k,l}, \\ 0 \le k \le N-1, 0 \le l \le M-1, \end{aligned} \tag{7}$$

where $f_{D,h}$ is the Doppler frequency and $\hat{\phi_h} = \phi_h - 2\pi \tau_h (f_c - \frac{B}{T}\tau_h)$ is the random phase of target h, respectively, and $\mathbf{W}$ is the AWGN matrix.

To detect angle information about the targets, we need to include the multi-antenna system in the radar system. Suppose that there are N_{T} transmit antennas and N_{R} receive antennas. Using the concept of the virtual antenna array, such system can be equivalent to the phased-array radar system with $Q = N_{\text{T}} N_{\text{R}}$ receive antennas. It follows that the received signal matrix of (7) can be extended as [40]:

$$\begin{aligned} (\mathbf{F}_i)_{k,l} = e^{-j2\pi f_c \mathbf{x}_i^H \mathbf{r}_h / c} \sum_{h=1}^{H} b_h e^{-j2\pi T f_{D,h} l} e^{-j2\phi \tau_h k \Delta f} e^{-j\hat{\phi_h}} \\ + (\mathbf{W}_i)_{k,l}, \quad 0 \le k \le N \\ - 1, 0 \le l \le M-1, 0 \le i \le Q-1, \end{aligned} \tag{8}$$

where $\mathbf{F}_i$ is the received signal matrix at virtual antenna $i+1$; $\mathbf{x}_i$ and $\mathbf{r}_h$ are the position vector of virtual antenna $i+1$ and direction vector of target h, respectively.

With (8), we basically can conduct the three-dimensional discrete Fourier transform (3D-DFT) to extract the range, Doppler velocity, and angle of targets, and then represent the result using the Range-Doppler-Angle (RDA) map as [39], [40]:

$$\begin{aligned} \left(\mathbf{M}^{\text{RDA}}\right)_{n+1,m+1,q+1} &= \text{DFT}_{\text{3D}}\left(\mathbf{F}\right)(n,m,q) \\ &= \sum_{i=0}^{Q_{\text{DFT}}-1} \sum_{k=0}^{N_{\text{DFT}}-1} \sum_{l=0}^{M_{\text{DFT}}-1} \left(\mathbf{F}_i\right)_{k,l} e^{\frac{j2\pi lm}{M_{\text{DFT}}}} e^{\frac{j2\pi kn}{N_{\text{DFT}}}} e^{\frac{j2\pi iq}{Q_{\text{DFT}}}}, \\ 0 \le n \le N_{\text{DFT}} - 1, &\ 0 \le m \le M_{\text{DFT}} - 1, 0 \le q \le Q_{\text{DFT}} - 1, \end{aligned} \tag{9}$$

where N_{DFT}, M_{DFT}, Q_{DFT} are the numbers of DFT in the frequency, time, and spatial domains, respectively. Note that $N_{\text{DFT}} = N$, $M_{\text{DFT}} = M$, and $Q_{\text{DFT}} = Q = N_{\text{T}} N_{\text{R}}$ are considered in this paper for simplicity.

B. RA Map and Point Cloud Generation

The RA map and point cloud are two main types of data that we use for object recognition in this paper. The point cloud can be directly obtained from the RDA map by using the constant false alarm rate (CFAR) detection [43], as a radar point in the point cloud is simply the detected target point with the associated range, Doppler velocity, angle, and the estimated signal-to-noise ratio (SNR). Differently, the RA map needs some further processing to obtain. Hence, we in this paper consider using the following process to obtain the RA map:

$$\left(\mathbf{M}^{\text{RA}}\right)_{n,q} = \max_{m} \left(\left| \left(\mathbf{M}^{\text{RDA}}\right)_{n,m,q} \right|^2 \right), \tag{10}$$

where $n = 1, \ldots, N$, $q = 1, \ldots, Q$. Note that although the above RA map and point cloud generations are standard, more advanced approaches might also be applied. As a final remark, although some advanced radar systems might be able to provide the 3D RA map and point cloud, to better illustrate our proposed approach in Section III, we assume that the RA map and the point

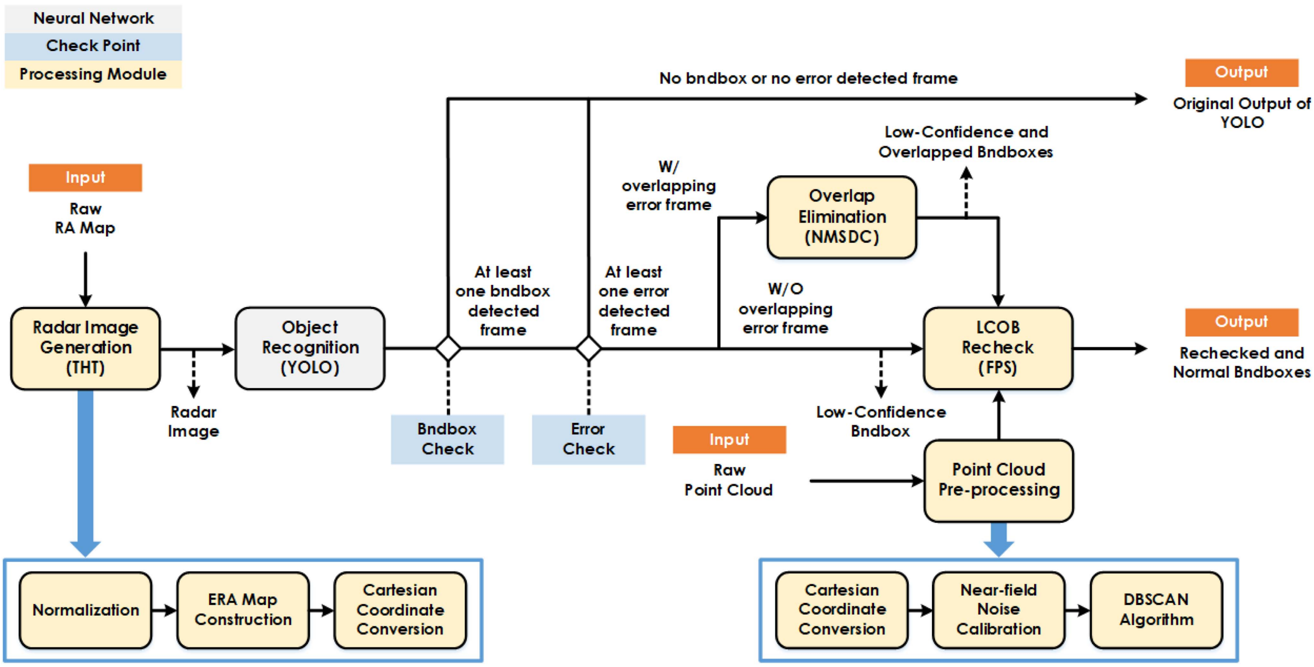


Fig. 1. Illustration of the proposed YOLO-ORE system.

cloud are with only the conventional two-dimensional (2D) RA map and point cloud for simplicity. As demonstrated in Section IV, The extension to considering the 3D RA map and/or point cloud is possible.[2]

III. Proposed YOLO-ORE System for Object Recognition With Radar

In this section, we present the proposed deep learning-aided object recognition approach for radar systems, namely, the YOLO-ORE system, where the system consists of a radar image generation module, a YOLO object recognition system, and an object recheck system. As illustrated in Fig. 1, the RA map obtained from the radar is first fed into the image generation module to generate the radar image for YOLO. Then, YOLO is used to determine the basic object recognition result. Since the result of YOLO might contain some errors, the recheck system, consisting of a point cloud pre-processing, an overlap elimination, and a low-confidence and overlapped bndbox (LCOB) Recheck, is used to modify the result obtained from YOLO and creates the final object recognition result. In the following, the YOLO-ORE system will be elaborated in detail. In addition, the frequently used notations of this paper are summarized in the table on the top of next page.

[2]Note that since the YOLO-based object recognition that will be introduced in Section III-B can only be used with the 2D radar image, the 3D RA map needs to be first compressed into a 2D RA map by certain approaches, e.g., the non-coherent combining of RA map [41], and then forwarded to the radar image generation proposed in the next section. If we want to use the 3D RA map without compression, we might need to replace the YOLO with other object recognition approaches that take the 3D data as input, e.g., [42].

A. Radar Image Generation

Since the RA map obtained from the radar detection is not an appropriate input for YOLO, we need to convert the RA map to the radar image that can be appropriately used by YOLO. To do this, we first generate the enhanced RA (ERA) map from the RA map by using the THT. Then, the RA map which is with the polar coordination would be converted to the Cartesian coordination, leading to the radar image that can be used as the input for the object recognition of YOLO.

Specifically, as the signal strength for RA bins in the RA map can have a wide range of variations, we need to conduct normalization and enhance the possible targets. To do this, we first define the maximum and minimum signal strength of all RA maps as:

$$S_{\max} = \max_{n,q,n_{\mathrm{f}}}\left(\left(\mathbf{M}^{\mathrm{RA}}\right)_{n,q}(n_{\mathrm{f}})\right), \quad S_{\min} = \min_{n,q,n_{\mathrm{f}}}\left(\left(\mathbf{M}^{\mathrm{RA}}\right)_{n,q}(n_{\mathrm{f}})\right),$$
$$n = 1,\ldots,N, q = 1,\ldots,Q, n_{\mathrm{f}} = 1,\ldots,N_{\mathrm{f}}, \tag{11}$$

where $(\mathbf{M}^{\mathrm{RA}})_{n,q}(n_{\mathrm{f}})$ is the RA map obtained at the n_{f}-th measurement frame and N_{f} is total number of measurement frames. With (11), we can acquire the normalized RA (NRA) map as follow:

$$\left(\mathbf{M}^{\mathrm{NRA}}\right)_{n,q}(n_{\mathrm{f}}) = \frac{\left(\mathbf{M}^{\mathrm{RA}}\right)_{n,q}(n_{\mathrm{f}}) - S_{\min}}{S_{\max} - S_{\min}},$$
$$n = 1,\ldots,N, q = 1,\ldots,Q, n_{\mathrm{f}} = 1,\ldots,N_{\mathrm{f}}, \tag{12}$$

Then, to enhance the possible targets, we let the upper bound for the strength be 1 and let the lower bound be the medium signal strength of all NRA maps collected during the measurement. Based on the upper and lower bounds, we adjust the signal strength of the ERA maps such that they can be fit into the upper and lower bounds. Specifically, the lower and upper bounds of the signal strength are given as:

$$S_{\mathrm{UB}} = 1, \quad S_{\mathrm{LB}} = \mathrm{medium}_{n,q,n_{\mathrm{f}}}\left(\left(\mathbf{M}^{\mathrm{NRA}}\right)_{n,q}(n_{\mathrm{f}})\right),$$
$$n = 1,\ldots,N, q = 1,\ldots,Q, n_{\mathrm{f}} = 1,\ldots,N_{\mathrm{f}}, \tag{13}$$

where $\mathrm{median}_{n,q,n_{\mathrm{f}}}((\mathbf{M}^{\mathrm{NRA}})_{n,q}(n_{\mathrm{f}}))$ is the median among all signal strength values of the NRA maps. Then, with the upper and lower bounds, the ERA map can be expressed as:

$$\left(\mathbf{M}^{\mathrm{ERA}}\right)_{n,q}(n_{\mathrm{f}})$$
$$= \begin{cases} \left(\left(\mathbf{M}^{\mathrm{NRA}}\right)_{n,q}(n_{\mathrm{f}})\right)^{x_e} & , \left(\mathbf{M}^{\mathrm{NRA}}\right)_{n,q}(n_{\mathrm{f}}) \geq S_{\mathrm{LB}} \\ 0, & \left(\mathbf{M}^{\mathrm{NRA}}\right)_{n,q}(n_{\mathrm{f}}) < S_{\mathrm{LB}} \end{cases},$$
$$n = 1,\ldots,N, q = 1,\ldots,Q, n_{\mathrm{f}} = 1,\ldots,N_{\mathrm{f}}, \tag{14}$$

where x_e is a parameter that controls the highlighting of the targets.

Given an ERA map whose representation adopts the range and angle as the x-axis and y-axis, respectively, we convert it to be with the Cartesian coordinate system, where the conversion is as simple as converting the Polar coordinate system to the Cartesian coordinate system. The converted ERA map is called RR map (or radar image), and the element of the RR map is called RR bin. Note that with the Cartesian coordinate system, the radar image in this paper is indeed a 2D image with 416×416 pixels; the exemplary illustration of the RA map, the generated radar image, and the corresponding real-world camera image are shown as in Fig. 2, where the figures only illustrate the region of interest in the measurements.

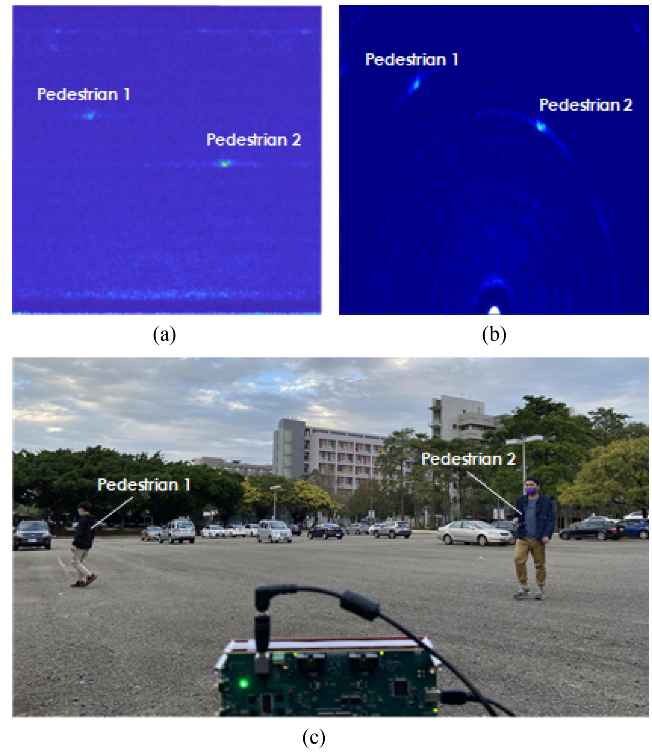


Fig. 2. An example of input and output of the radar image generation and the corresponding camera image. (a) Raw RA map. (b) Radar image. (c) Corresponding camera image.

B. YOLO-Based Object Recognition

With the generated radar image, we then use YOLO to conduct the basic object recognition. YOLO is one of the most famous one-stage NN models for object recognition [29]. Conventionally, completing an object recognition task requires two steps, namely, object detection and object classification. The former one is to determine whether the objects exist and the locations of objects; the latter one is to classify the objects into different classes. Different from the two-stage approaches, YOLO performs the object detection and object classification simultaneously in one-shot, which facilitates the implementation speed. In the following, we briefly introduce the implementation of YOLO. Then, we point out the issues of YOLO that motivates the object recheck system proposed in our YOLO-ORE object recognition approach.

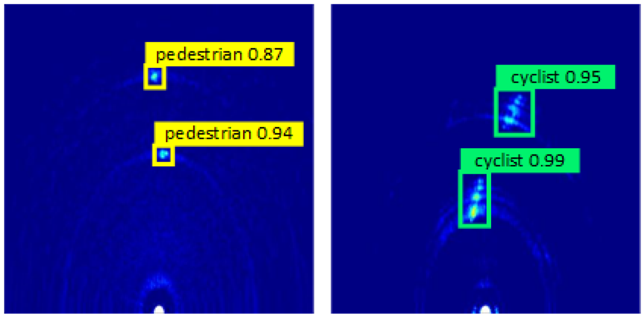
1) Introduction of YOLO: Briefly speaking, YOLO first divides the input image into $S \times S$ grid cells, and then, for each grid cell, it predicts B bndboxes with the corresponding object confidence that represents how confident YOLO thinks about whether the bndbox contains an object. Then, for each bndbox, there are 6 resulting predictions: x, y, l, w, C, and Class, where x and y indicate the centering coordinate of the bndbox; l and w are the length and width of the bndbox, respectively; and Class and C are the class of the object in the bndbox and the corresponding confidence level, respectively. Finally, only the bndboxes with a confidence level higher than a given threshold would be appended to the input image. Note that there are two main thresholds for YOLO: confidence threshold $T_{\mathrm{C,YOLO}}$ and intersection over union (IOU) threshold $T_{\mathrm{IOU,YOLO}}$, respectively. The former is to eliminate the bndboxes lower than the threshold and the latter is to remove the bndboxes with the same predicted classes overlapped with one another whose IOU is higher than the threshold. The IOU of bndboxes of two objects i and j is given as:

$$IOU_{ij} = \frac{A_{ij}^{\mathrm{inter}}}{A_{ij}^{\mathrm{union}}}, \tag{15}$$

where the A_{ij}^{inter} and A_{ij}^{union} are the size of intersection and union of bndboxes of two objects i and j, respectively. The exemplary output images of YOLO are shown in Fig. 3.

Since YOLO has been investigated for many years, there exist multiple versions of YOLO. Since YOLOv3 [29] has been mature and shown stable in many conditions, we in this paper adopt YOLOv3 in our proposed YOLO-ORE system. Nevertheless, depending on the system requirements, e.g., small system size and/or low power consumption, and applications, other versions of YOLO can also be used. We note that as YOLO is a supervised machine learning model, it needs to be well-trained for the use of the object recognition.

2) Object Overlapping and Misclassification Errors of YOLO: Although a well-trained YOLO can conduct efficient



(a) (b)

Fig. 3. The exemplary output images of YOLO: (a) The result consisting of two pedestrians; (b) The result consisting of two cyclists.

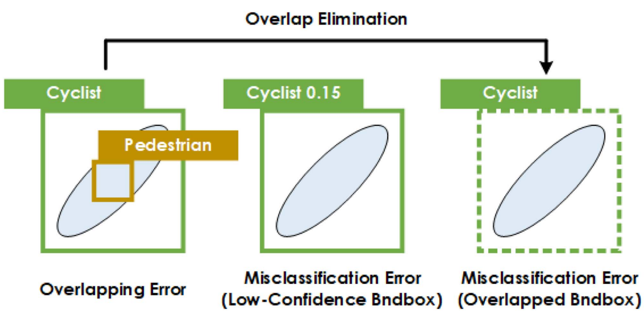
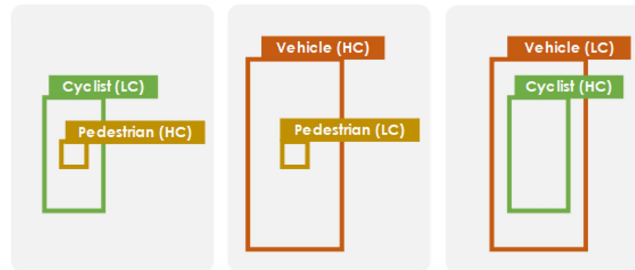


Fig. 4. The examples of overlapping and misclassification errors.

object recognition based on radar images, it still has some issues to resolve. Specifically, we observe that mainly two types of errors exist in our object recognition with radar application: overlapping error and misclassification error. The overlapping error comes from that in certain situations, YOLO could create multiple bndboxes with different predicted classes overlapping with one another. However, since different objects cannot physically overlap with one another at the same position, this type of overlapping is mostly erroneous. Note that although YOLO can effectively resolve the overlapping between objects of the same class, it does not take much care of the overlapping between objects with different classes. The misclassification error comes from that the predicted bndboxes could have very low confidence or that the bndboxes are originally overlapped with other bndboxes, and this type of issue is called misclassification error because it commonly leads to the misclassification of the object. We note that the latter case of the misclassification error is created by that resolving the overlapping error via using the overlap elimination in our proposed YOLO-ORE. In addition, although the very-low-confidence bndboxes for the misclassification error might be eliminated by increasing $T_{\mathrm{C,YOLO}}$ for the object recognition, this also increases the possibility that we miss the object, and thus eliminates the chances that we can re-classify the object. The illustrations of the errors are provided in Fig. 4.

To resolve the overlapping and misclassification errors, in the YOLO-ORE system, we propose using additional processing modules, namely, the overlap elimination and LCOB recheck, based on the point cloud. The relevant discussions of these two modules are included in the next three subsections.



(a) (b) (c)

Fig. 5. Illustration of overlapping errors, where the HC indicates the high confidence level and LC indicates the low confidence level. (a) Case 1. (b) Case 2. (c) Case 3.

C. Point Cloud Pre-Processing

To resolve the overlapping and misclassification errors of YOLO, we conduct the rechecking by using the point cloud generated by the radar system. Specifically, after obtaining the object recognition result of YOLO, we then check whether there exist overlapping and misclassification errors by checking whether there exist objects whose IOU ratio of their corresponding bndboxes is larger than zero and by checking whether there exist objects whose confidence level is lower than the low-confidence threshold for ORE system $T_{\mathrm{LC,ORE}}$, respectively. If so, we then initialize the point cloud pre-processing which prepares the later object rechecking.

The point cloud pre-processing mainly consists of three steps. Since the generated point cloud by the radar system might not be with the Cartesian coordinate system, we first convert the point cloud into the same Cartesian coordinate system as the radar image used by YOLO. Then, because there might exist certain noisy points in the very near-field of the radar system, we remove them from the point cloud according to the calibration. Finally, to cluster the points into objects, we conduct the density-based spatial clustering of applications with noise (DBSCAN) algorithm that is widely used for the clustering of points in the point cloud. The clustered point cloud would then be sent to the proposed object rechecking system that is elaborated in Section III-E.

D. Overlap Elimination

When there exist overlapping errors, as illustrated in Fig. 5, the overlap elimination is used to resolve them. Specifically, although YOLO already uses the non-maximum suppression (NMS) algorithm to avoid the overlapping of bndboxes for the objects of the same class [44], it does not avoid the overlapping for the objects with different classes. To solve this problem, we use the proposed non-maximum suppression between different classes (NMSDC) algorithm, which is a modified version of the NMS algorithm, to eliminate the overlapping of bndboxes of different classes.

Different from the NMS algorithm that uses the IOU of objects to deal with the overlapping issue, the NMSDC algorithm considers using the intersection over bndbox (IOB) to conduct the overlap elimination. The IOB of bndboxes of two objects i

Algorithm 1: NMSDC Algorithm.

1: **Input** S_{detBox} : Detected bndbox set from YOLO
2: $S_{\text{NMSDC}} \leftarrow \emptyset$ ▷Initialize empty set
3: $S_{\text{NMSDC}} \leftarrow$ Sorted S_{detBox} by confidence levels
4: **for** $i = 1, 2, \ldots, |S_{\text{NMSDC}}|$ **do**
5: $b_i \leftarrow i$-th box in S_{detBox}
6: **if** confidence of level of $b_i \neq 0$ **then**
7: **for** $j = i + 1, \ldots,$ **do**
8: $b_j \leftarrow j$-th box in S_{detBox}
9: **if** confidence of $b_j \neq 0$ **then**
10: Compute IOB_{ij} of b_i and b_j
11: **if** $IOB_{ij} > T_{\text{IOB,NMSDC}}$ **then** ▷IOB test
12: Set confidence of b_j equal to 0.
13: **end if**
14: **end if**
15: **end for**
16: $S_{\text{NMSDC}} \leftarrow S_{\text{NMSDC}} \cup b_i$
17: **end if**
18: **end for**
19: **Output** S_{NMSDC} : Bndbox set after NMSDC

and j is defined as:

$$IOB_{ij} = \max\left(IOB_{ij}^{i}, IOB_{ij}^{j}\right), \tag{16}$$

where IOB_{ij}^{i} is the IOB of object i given by

$$IOB_{ij}^{i} = \frac{A_{ij}^{\text{inter}}}{A_i} \tag{17}$$

and A_i is the size of bndbox of object i. We note that the use of IOB instead of IOU is because IOU can be significantly biased when two objects have very different sizes of bndboxes, as using IOU can lead to the issue that the objects with very different sizes but significantly overlapped with one another might not be easily detected regarding their overlapping (IOU is small); the illustrations are provided in Fig. 5.

Knowing how to compute the IOB of objects, we can then conduct the overlap elimination by using the proposed NMSDC algorithm. Suppose we have K_{OVL} objects to overlap with one another, we first sort them by their confidence levels in descending order. Then, starting with the object with the largest confidence level, we compute its IOBs with all other objects, respectively. Then, whenever there exists a IOB that is larger than the given IOB threshold for NMSDC $T_{\text{IOB,NMSDC}}$, we eliminate the one with the lower confidence level among the two objects associated with this IOB. This process then proceeds from the objects with higher confidence levels to those with lower confidence levels, and then stops when all objects are checked and/or eliminated. The overall procedure of the proposed NMSDC algorithm is summarized in Algorithm 1, where Step 12 indicates that those bndbox set to be with confidence equal to zero will no longer be tested and included in the final bndbex set.

E. Low-Confidence and Overlapped Bndbox Recheck

After the overlap elimination, the resulting object recognition results would then be fed into the LCOB Recheck for the final

Fig. 6. The block diagram of LCOB Recheck.

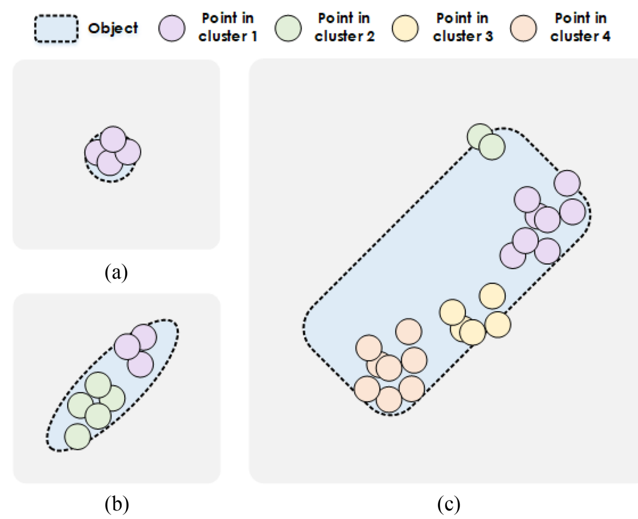


Fig. 7. Examples of clusters of radar ppoints after the point cloud pre-processing, where (a), (b), and (c) are clusters of pedestrian, cyclist, and vehicle, respectively.

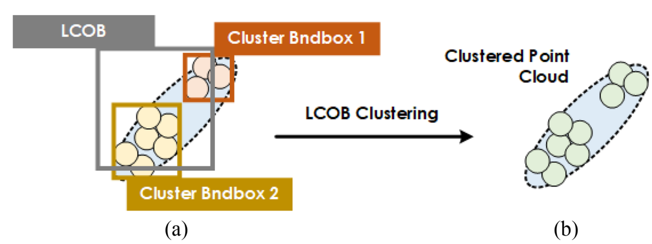


Fig. 8. Examples of the LCOB clustering, where two clusters are associated to an object.

object recheck. The LCOB recheck is a module mainly used to deal with the misclassification errors via using the features-of-point-cloud based SVM (FPS) classifier to re-classify the objects. Note that since we only re-classify the low-confidence bndboxes and those bndboxes that were overlapped with other bndboxes and resolved by the overlap elimination, we only input those bndboxes along with their corresponding point cloud clusters to the LCOB recheck. We then recheck the objects by using FPS with their point cloud features. As Illustrated in Fig. 6, the LCOB recheck consists of three blocks: the LCOB clustering, the feature extraction, and the FPS classifier. In the following, we will discuss them in detail.

1) LCOB Clustering: After point cloud pre-processing, we could have multiple clusters in the point cloud, where each cluster contains multiple radar points. However, because the sizes of different objects could be very different, the numbers of points and their relevant location distribution of objects could be significantly different. The examples of clusters of pedestrians,

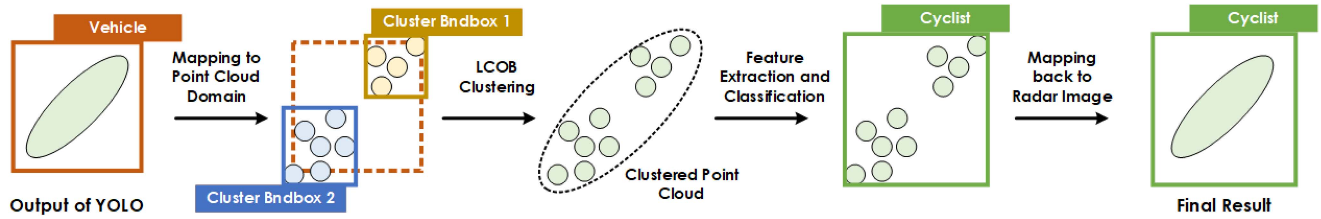


Fig. 9. An example of the LCOB Recheck.

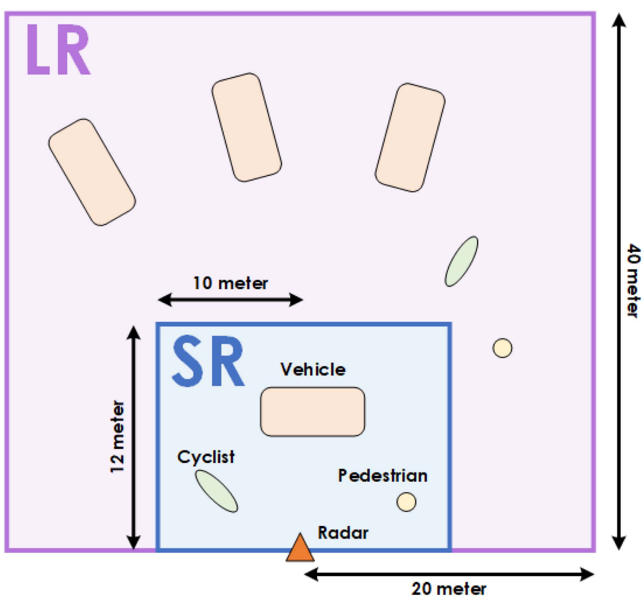


Fig. 10. Illustration of the short range (SC) and long range (LR) measurement scenarios.

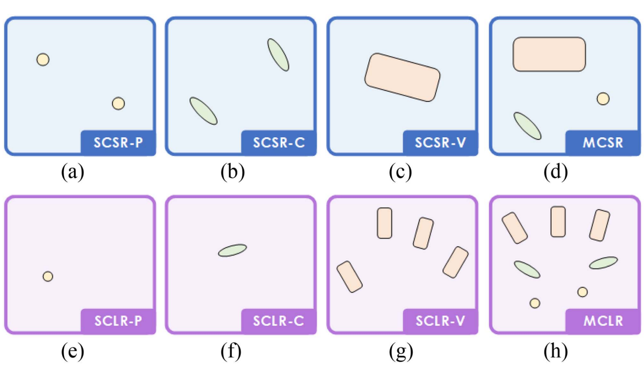


Fig. 11. Illustrations of different measurement cases. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3. (d) Scenario 4. (e) Scenario 5. (f) Scenario 6. (g) Scenario 7. (h) Scenario 8.

cyclists, and vehicles after the point cloud pre-processing are illustrated in Fig. 7.

Seeing that an object could be composed of multiple clusters of radar points, we need to associate different clusters to objects recognized by YOLO and conduct the recheck. To do this, we leverage the idea of bndbox in YOLO. Specifically, for each cluster in the point cloud, we first define its cluster bndbox by using:

$$x_{\mathrm{CB}} = \frac{X_{\mathrm{CB}}^{\max} + X_{\mathrm{CB}}^{\min}}{2} \quad ; \quad y_{\mathrm{CB}} = \frac{Y_{\mathrm{CB}}^{\max} + Y_{\mathrm{CB}}^{\min}}{2},$$
$$l_{\mathrm{CB}} = X_{\mathrm{CB}}^{\max} - X_{\mathrm{CB}}^{\min} \quad ; \quad w_{\mathrm{CB}} = Y_{\mathrm{CB}}^{\max} - Y_{\mathrm{CB}}^{\min}, \tag{18}$$

where $X_{\mathrm{CB}}^{\max}$, $Y_{\mathrm{CB}}^{\max}$ are the x-axis and y-axis values of the rightmost and top-most radar points of the cluster; $X_{\mathrm{CB}}^{\min}$, $Y_{\mathrm{CB}}^{\min}$ are the x-axis and y-axis values of the leftmost and bottom-most radar points of the cluster; $(x_{\mathrm{CB}}, y_{\mathrm{CB}})$ is the centering location of the cluster bndbox on the Cartesian coordinate system; and w_{CB} and l_{CB} are the width and length of the cluster bndbox, respectively.

With the cluster bndboxes of different clusters, we then use the IOB between the cluster bndboxes and the bndboxes of objects recognized by YOLO to conduct the cluster-object association. To do this, for a pair of cluster bndbox i and object bndbox j, we compute its IOB and see whether the corresponding IOB value is larger than the IOB threshold $T_{\mathrm{IOB,LCOBC}}$. If yes, then cluster i is associated to object j. Such association process is then conducted for every pair of cluster bndbox and object bndbox. Note that there could exist clusters that are not associated with any object. In this case, we remove them from the point cloud. In addition, there could also exist the case that a cluster is associated with multiple objects. In this case, we only associate the cluster with the object that has the largest IOB. Since the above procedure can construct new clusters of radar points for objects, it is called LCOB clustering. An example of the LCOB clustering is illustrated in Fig. 8.

2) Feature Extraction: The LCOB clustering helps us associate clusters with the objects. To conduct rechecks for objects of interest, we aim to use a FPS classifier. Since the FPS classifier can perform well only if features of the clustered radar points of objects are well-acquired, we here propose a feature extraction approach.

Specifically, to re-classify an object, we consider using three types of features of the clustered radar points of the object, namely, the shape, intensity, and velocity of the object. The shape-based features are used to represent the physical geometry of the clustered radar points of the object. Thus, the basic shape-based features here include the number of radar points in the object of interest n_{CB} as well as the length L_{CB}, the width W_{CB}, and the shape-volume S_{CB} of the clustered radar points associated with the object of interest:

$$n_{\mathrm{CB}} = \text{number of points associated to the objects;}$$
$$L_{\mathrm{CB}} = X_{\mathrm{CB}}^{\max} - X_{\mathrm{CB}}^{\min}, W_{\mathrm{CB}} = Y_{\mathrm{CB}}^{\max} - Y_{\mathrm{CB}}^{\min};$$

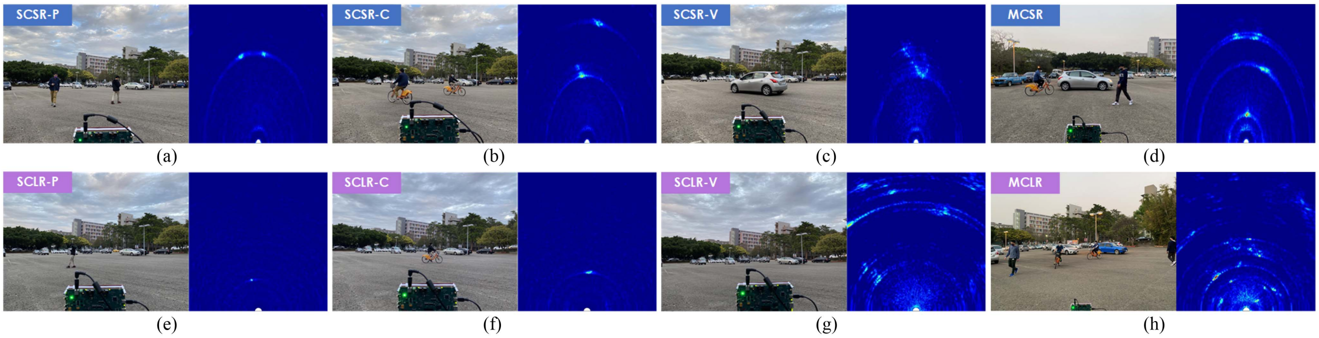


Fig. 12. Exemplary radar images and corresponding camera images for each scenario. (a) Scenario 1. (b) Scenario 2. (c) Scenario 3. (d) Scenario 4. (e) Scenario 5. (f) Scenario 6. (g) Scenario 7. (h) Scenario 8.

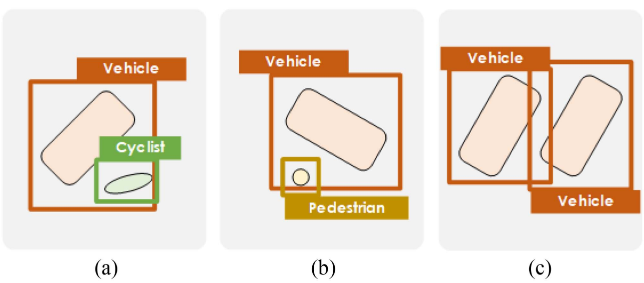


Fig. 13. Illustrations of correct overlappings. (a) Case 1. (b) Case 2. (c) Case 3.

$$S_{\mathrm{CB}} = L_{\mathrm{CB}} \times W_{\mathrm{CB}}. \tag{19}$$

Note that different from the length and width in (18) that only considers the radar points in a cluster bndbox, the length and width here consider all radar points associated to the object. To obtain additional shape-based features of the object, we further consider the standard deviations of the x-axis and y-axis values for all points in the object, given by:

$$X_{\mathrm{CB}}^{\mathrm{SD}} = \sqrt{\frac{1}{n_{\mathrm{CB}}} \sum\nolimits_{i=1}^{n_{\mathrm{CB}}} (x_i - \overline{x})^2}, \quad Y_{\mathrm{CB}}^{\mathrm{SD}} = \sqrt{\frac{1}{n_{\mathrm{CB}}} \sum\nolimits_{i=1}^{n_{\mathrm{CB}}} (y_i - \overline{y})^2}, \tag{20}$$

where x_i and y_i are the x-axis and y-axis values of radar point i in the object, respectively; and $\overline{x} = \frac{1}{n_{\mathrm{CB}}} \sum_{i=1}^{n_{\mathrm{CB}}} x_i$ and $\overline{y} = \frac{1}{n_{\mathrm{CB}}} \sum_{i=1}^{n_{\mathrm{CB}}} y_i$. The intensity-based features are used to represent the target reflection effect. We thus consider

$$D_{\mathrm{CB}} = \frac{n_{\mathrm{CB}}}{S_{\mathrm{CB}}}, \quad \mathrm{SNR}_{\mathrm{CB}} = \max_i \mathrm{SNR}_i, \tag{21}$$

as the intensity feature of the object, where SNR_i is the estimated SNR given by the radar system for the radar point i of the object. Finally, the velocity-based features represent the movement of the object. We thus consider the average velocity $\overline{V}_{\mathrm{CB}}$, maximum difference velocity $V_{\mathrm{CB}}^{\mathrm{Diff}}$, and the velocity standard deviation $V_{\mathrm{CB}}^{\mathrm{SD}}$ as the velocity features:

$$\overline{V}_{\mathrm{CB}} = \frac{1}{n_{\mathrm{CB}}} \sum_{i=1}^{n_{\mathrm{CB}}} v_i, V_{\mathrm{CB}}^{\mathrm{Diff}} = V_{\mathrm{CB}}^{\max} - V_{\mathrm{CB}}^{\min},$$

TABLE I
CONFIGURATIONS FOR MEASUREMENTS

Configuration	**SR**	**LR**
Number of Transmit Antenna	12	12
Number of Receive Antenna	16	16
Starting Carrier Frequency (GHz)	77	77
Frequency Slope (MHz/us)	70	45
Number of ADC Samples	256	256
ADC Sampling Frequency (Msps)	8	15
Bandwidth (MHz)	2380	810
Maximum Range (m)	16	46
Range Resolution (cm)	6.90	19.83
Azimuth Angle Resolution (degree)	1.4	1.4
Elevation Angle Resolution (degree)	18	18

TABLE II
CONSTITUTIONS OF DATASETS FOR STANDARD SCENARIOS

SCSR	Total	Ped.	Cyc.	Veh.	Scenario
Image	4000	1000	1000	2000	1-3
Ground Truth	6000	2000	2000	2000	
SCLR	Total	Ped.	Cyc.	Veh.	Scenario
Image	4310	2000	2000	310	5-7
Ground Truth	6020	2000	2000	2020	
SCMR	Total	Ped.	Cyc.	Veh.	Scenario
Image	8310	3000	3000	2310	1-3,5-7
Ground Truth	12020	4000	4000	4020	
MCSR	Total	Ped.	Cyc.	Veh.	Scenario
Image	2416	-	-	-	4
Ground Truth	7248	2416	2416	2416	
MCLR	Total	Ped.	Cyc.	Veh.	Scenario
Image	824	-	-	-	8
Ground Truth	6550	1648	1648	3254	
MCMR	Total	Ped.	Cyc.	Veh.	Scenario
Image	3240	-	-	-	4,8
Ground Truth	13798	4064	4064	5670	

TABLE III
CONFIGURATION OF YOLO

Parameter	**Value**
Classes	3
Filters	24
Epoch	300
Batchsize	10
Width	416 (pixel)
Height	416 (pixel)
Channel	3 (RGB)
Learning Rate	0.001
Momentum	0.9
Decay	0.0005

TABLE IV
THRESHOLD SETS OF THE YOLO-ORE SYSTEM

Threshold	**SC**	**MC**
$T_{\text{C,YOLO}}$	0.1	0.1
$T_{\text{IOU,YOLO}}$	0.25	0.25
$T_{\text{LC,ORE}}$	0.5	0.25
$T_{\text{IOB,NMSDC}}$	0.25	0.5
$T_{\text{IOB,LCOBC}}$	0.5	0.25

$$V_{\text{CB}}^{\text{SD}} = \sqrt{\frac{1}{n_{\text{CB}}} \sum\nolimits_{i=1}^{n_{\text{CB}}} (v_i - \overline{V}_{\text{CB}})^2}, \tag{22}$$

where v_i is the Doppler velocity for radar points i of the object; $V_{\text{CB}}^{\max}$ and $V_{\text{CB}}^{\max}$ are the maximum and minimum velocities of the radar points of the object, respectively. The above features would then be used as the input features for the FPS classifier to conduct the re-classification.

3) FPS Classifier: SVM and its variants have been widely used for different classification problems. Thus, after the feature extraction, we aim to use the extracted features along with the FPS classifier [22] to realize the nonlinear classification which provides re-classifications for the objects that need to be rechecked. Combining the LCOB clustering, the feature extraction, and the FPS classifier, a processing example of the complete LCOB recheck is illustrated in Fig. 9.

IV. EXPERIMENTAL RESULTS

In this section, extensive experimental results of the proposed object recognition approach are provided to validate the performance of the proposed YOLO-ORE system.

A. Data Collection and Dataset Construction

In the experiments, we use a AWR2243 cascade radar produced by Texas Instruments (TI) [41]. The AWR2243 cascade radar is an integrated FMCW radar consisting of 12 transmit antennas and 16 receive antennas (12T16R). The AWR2243 cascade radar is capable of operating between the 76 to 81 GHz frequency band. With an appropriate configuration, the AWR2243 cascade radar basically can achieve the 150 m maximum range with the 60 cm range resolution. Besides, because it has 3 transmit antennas placed in the vertical direction, it can generate the 3D point cloud with the azimuth and elevation angle resolution being 1.4° and 18°, respectively.

In our experiments, we mount the radar sensor on a tripod so that the radar is 90 cm above the ground for measurement. We consider several types of measurement datasets, distinguished by their numbers of classes and the maximum distance between the objects and the radar. Specifically, we denote the measurements consisting of only a single class of objects as single class (SC) measurements and denote the measurements consisting of more than a single class of objects as multiple class (MC) measurements. Also, we consider different maximum ranges of the measurements, denoted as short range (SR), long range (LR), and mixed range (MR) measurements. The SR measurement considers that the objects are located within a 12×20 m area, while the LR measurement considers that the objects are located within a 40×40 m area and most of the objects are outside the SR area. The MR measurement is the combination of data measured in SR and LR. The illustration of the SC and LR measurement scenarios in this paper is shown in Fig. 10. The configurations of the radar for SR and LR are shown in Table I.

By the above descriptions, we can obtain six types of scenarios for our measurements: SCSR, SCLR, SCMR, MCSR, MCLR, and MCMR, and they are considered as standard scenarios in the paper. We then consider three different types of objects, namely, the pedestrian, cyclist, and vehicle, for the experiments. The illustrations of different measurement cases are shown in Fig. 11; the exemplary radar images and the corresponding camera images are shown in Fig. 12; and the constitutions of the datasets after measurements are as shown in Table II, where "Image" indicates the total number of radar images collected by the measurements and "Ground Truth" indicates the total number of objects in the collected radar images.

B. YOLO-ORE Training and Testing

To operate the YOLO-ORE system, we need to train two machines: the YOLO and the FPS. To configure the YOLO used in the experiments, we first consider the parameters as being provided in Table III, and then mostly follow the standard approach as recommended in [29] by the YOLO developers. The only differences are that the classes and filters are modified so that they can fit our datasets and that we set the epoch to 300. To configure the FPS, we consider the polynomial kernel with the order being 3. During the training and testing of the YOLO-ORE, we split the collected radar image data into training, validation, and testing datasets with the proportion of 72%, 18%, and 10% of all data samples, respectively. We use the K-fold cross-validation method during the training and set $K = 10$. The threshold sets used by the YOLO-ORE system in the experiments are provided in Table IV.

C. Experiment Results for Standard Scenarios

In this subsection, we evaluate the proposed YOLO-ORE objection recognition system in the standard scenarios described in Section IV-A. We note that because the radar detection is based on the line-of-sight (LoS) assumption, we only consider those objects that have LoS path when conducting the evaluation. We consider using Precision, Recall, and Accuracy for evaluation, and their definitions are as follows:

$$\text{Precision} = \frac{TP}{TP + FP};$$
$$\text{Recall} = \frac{TP}{TP + FN};$$
$$\text{Accuracy} = \frac{TP}{TP + FP + FN}, \tag{23}$$

where TP is the number of true positives, FP is the number of false positives, and FN is the number of false negatives. Note that here the accuracy is different from the conventional one because when evaluating the object detection performance, it is meaningless to check the number of the true negatives as the number of true negatives could be overwhelmingly large [45]. Note that a true negative means that the object does not exist and

TABLE V
EXPERIMENT RESULTS OF THE YOLO-ORE SYSTEM

Dataset	SCSR				SCLR				SCMR			
Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
TP	586	236	202	148	560	204	183	173	1081	353	387	341
FP	8	6	1	1	86	19	14	53	97	34	30	33
FN	36	2	4	30	27	5	12	10	110	19	23	68
Precision	98.65%	97.52%	99.51%	99.33%	86.96%	91.48%	92.89%	76.55%	91.77%	91.21%	92.81%	91.18%
Recall	94.21%	99.16%	98.06%	83.15%	95.40%	97.61%	93.85%	94.54%	90.76%	94.89%	94.39%	83.37%
Accuracy	**93.02%**	96.72%	97.58%	82.68%	**83.21%**	89.47%	87.56%	73.31%	**83.93%**	86.95%	87.95%	77.15%
Dataset	MCSR				MCLR				MCMR			
Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
TP	638	216	197	225	637	151	156	330	1317	370	367	580
FP	71	29	15	27	74	25	19	30	137	74	27	36
FN	85	25	44	16	21	13	8	0	100	43	46	11
Precision	89.99%	88.16%	92.92%	89.29%	89.59%	85.80%	89.14%	91.67%	90.58%	83.33%	93.15%	94.16%
Recall	88.24%	89.63%	81.74%	93.36%	96.81%	92.07%	95.12%	100%	92.94%	89.59%	88.86%	98.14%
Accuracy	**80.35%**	80.00%	76.95%	83.96%	**87.02%**	79.89%	85.25%	91.67%	**84.75%**	75.98%	83.41%	92.50%

TABLE VI
EXPERIMENT RESULTS OF THE YOLO-SVMV2 SYSTEM

Dataset	SCSR				SCLR				SCMR			
Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
TP	589	236	205	148	571	207	189	175	1100	360	393	347
FP	23	21	2	0	132	46	27	59	204	65	103	36
FN	33	2	1	30	16	2	6	8	91	12	17	62
Precision	96.24%	91.83%	99.03%	100%	81.22%	81.82%	87.50%	74.79%	84.36%	84.71%	79.23%	90.60%
Recall	94.69%	99.16%	99.51%	83.15%	97.27%	99.04%	96.92%	95.63%	92.36%	96.77%	95.85%	84.84%
Accuracy	**91.32%**	91.12%	98.56%	83.15%	**79.42%**	81.18%	85.14%	72.31%	**78.85%**	82.38%	76.61%	77.98%
Dataset	MCSR				MCLR				MCMR			
Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
TP	660	221	208	231	645	159	152	330	1335	378	375	582
FP	116	49	24	43	99	37	32	30	222	139	45	38
FN	63	20	33	10	13	5	8	0	82	35	38	9
Precision	85.05%	81.85%	89.66%	84.31%	86.69%	81.12%	82.98%	91.67%	85.74%	73.11%	89.29%	93.87%
Recall	91.29%	91.70%	86.31%	95.85%	98.02%	96.95%	95.12%	100%	94.21%	91.53%	90.80%	98.48%
Accuracy	**78.67%**	76.21%	78.49%	81.34%	**85.20%**	79.10%	79.59%	91.67%	**81.45%**	68.48%	81.88%	92.53%

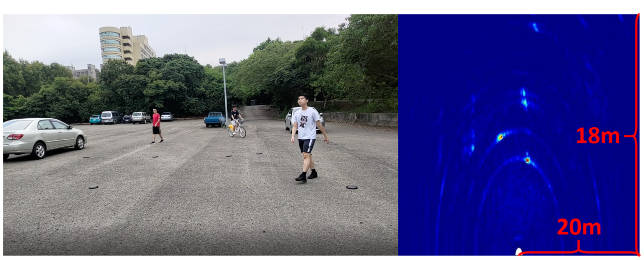


Fig. 14. An exemplary radar image and the corresponding camera image of NYCU parking lot scenario.

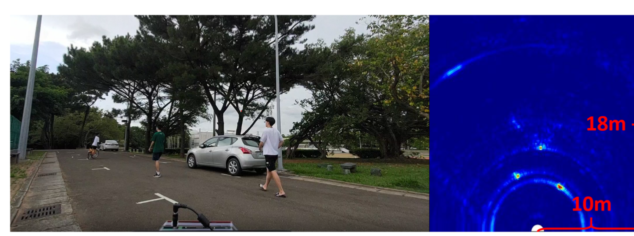


Fig. 15. An exemplary radar image and the corresponding camera image of NYCU road-side scenario.

the system says so. Therefore, as it could happen that most area of the radar image does not have objects, it becomes meaningless to include the true negatives in the performance evaluation metrics.

In this subsection, to provide a more comprehensive evaluation, we compare our proposed YOLO-ORE system with an approach modified from the approach in [37]. Note that the modification is because the approach proposed in [37] can only be applied to the two-object detection. Therefore, we modify the SVM part in [37] such that it can be used for object detection of more than two objects. To do the modification, we note that the original YOLO-SVM system in [37] let both YOLO and SVM conduct classifications, and then if empirically SVM can provide

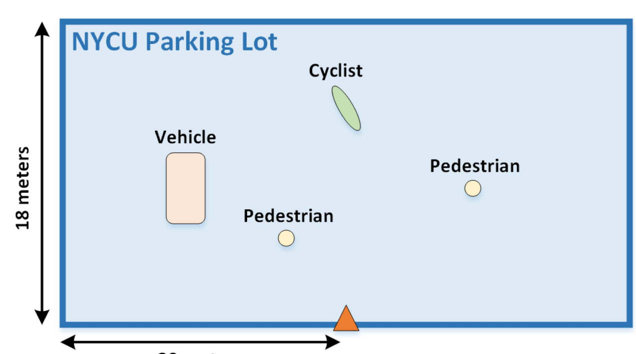


Fig. 16. Illustration of the NYCU parking lot scenario.

TABLE VII
COMPARISONS BETWEEN THE YOLO-ORE AND YOLO-SVMv2 SYSTEMS IN STANDARD SCENARIOS

Dataset		SCSR				SCLR				SCMR			
Acc.	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
	Y.-O.	93.02%	96.72%	97.58%	82.68%	83.21%	89.47%	87.56%	73.31%	83.93%	86.95%	87.95%	77.15%
	Y.-S.	91.32%	91.12%	98.56%	83.15%	79.42%	81.18%	85.14%	72.31%	78.85%	82.38%	76.61%	77.98%
	Incre.	**1.70%**	**5.60%**	-0.98%	-0.47%	**3.79%**	**8.29%**	**2.42%**	**1.00%**	**5.08%**	**4.57%**	**11.34%**	-0.83%
Rec.	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
	Y.-O.	94.21%	99.16%	98.06%	83.15%	95.40%	97.61%	93.85%	94.54%	90.76%	94.89%	94.39%	83.37%
	Y.-S.	94.69%	99.16%	99.51%	83.15%	97.27%	99.04%	96.92%	95.63%	92.36%	96.77%	95.85%	84.84%
	Incre.	-0.48%	0%	-1.45%	0%	-1.87%	-1.43%	-3.07%	-1.09%	-1.6%	-1.88%	-1.46%	-1.47%
Pre.	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
	Y.-O.	98.65%	97.52%	99.51%	99.33%	86.96%	91.48%	92.89%	76.55%	91.77%	91.21%	92.81%	91.18%
	Y.-S.	96.24%	91.83%	99.03%	100%	81.22%	81.82%	87.50%	74.79%	84.36%	84.71%	79.23%	90.60%
	Incre.	**2.39%**	**5.69%**	0.48%	-0.67%	**5.74%**	**9.66%**	**5.39%**	**1.76%**	**7.41%**	**6.5%**	**13.58%**	**0.58%**
Dataset		**MCSR**				**MCLR**				**MCMR**			
Acc.	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
	Y.-O.	80.35%	80.00%	76.95%	83.96%	87.02%	79.89%	85.25%	91.67%	84.75%	75.98%	83.41%	92.50%
	Y.-S.	78.67%	76.21%	78.49%	81.34%	85.20%	79.10%	79.59%	91.67%	81.45%	68.48%	81.88%	92.53%
	Incre.	**1.68%**	**3.79%**	-1.54%	**2.62%**	**1.82%**	**0.79%**	**5.66%**	0%	**3.3%**	**7.5%**	**1.53%**	-0.03%
Rec.	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
	Y.-O.	88.24%	89.63%	81.74%	93.36%	96.81%	92.07%	95.12%	100%	92.94%	89.59%	88.86%	98.14%
	Y.-S.	91.29%	91.70%	86.31%	95.85%	98.02%	96.95%	95.12%	100%	94.21%	91.53%	90.80%	98.48%
	Incre.	-3.05%	-2.07%	-4.57%	-2.49%	-1.21%	-4.88%	0%	0%	-1.27%	-1.94%	-1.94%	-0.34%
Pre.	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
	Y.-O.	89.99%	88.16%	92.92%	89.29%	89.59%	85.80%	89.14%	91.67%	90.58%	83.33%	93.15%	94.16%
	Y.-S.	85.05%	81.85%	89.66%	84.31%	86.69%	81.12%	82.98%	91.67%	85.74%	73.11%	89.29%	93.87%
	Incre.	**4.94%**	**6.31%**	**3.26%**	**4.98%**	**2.9%**	**4.68%**	**6.16%**	0%	**4.84%**	**10.22%**	**3.86%**	**0.29%**

TABLE VIII
OVERLAP ELIMINATION PERFORMANCE OF THE PROPOSED NMSDC ALGORITHM

Dataset	**SCSR**	**SCLR**	**SCMR**	**MCSR**	**MCLR**	**MCMR**
Number of Overlap before NMSDC	18	67	144	89	34	114
Number of Overlap after NMSDC	0	15	17	18	1	14
Number of Eliminated Overlap	18	52	127	71	33	100
Overlap Elimination Ratio	**100%**	**77.61%**	**88.19%**	**79.78%**	**97.06%**	**87.72%**

TABLE IX
FPS RE-CLASSIFICATION PERFORMANCE

Dataset	**SCSR**			**SCLR**			**SCMR**		
Class	Ped.	Cyc.	Veh.	Ped.	Cyc.	Veh.	Ped.	Cyc.	Veh.
Number of Correct Prediction	220	189	178	189	170	176	339	360	396
Total Prediction	223	192	178	201	183	177	347	381	403
Classification Accuracy	98.65%	98.44%	100%	93.53%	92.90%	99.44%	97.69%	94.49%	98.26%
Total Classification Accuracy	**98.99%**			**95.19%**			**96.82%**		
Dataset	**MCSR**			**MCLR**			**MCMR**		
Class	Ped.	Cyc.	Veh.	Ped.	Cyc.	Veh.	Ped.	Cyc.	Veh.
Number of Correct Prediction	188	192	230	106	124	300	312	328	545
Total Prediction	194	210	241	118	142	304	329	360	554
Classification Accuracy	96.91%	91.43%	95.44%	89.83%	87.32%	98.68%	94.83%	91.11%	98.38%
Total Classification Accuracy	**94.57%**			**93.97%**			**95.33%**		

TABLE X
COMPARISONS BETWEEN YOLO-ORE AND YOLO-SVMv2 SYSTEMS IN EXTENDED SCENARIOS

Dataset		**NYCU road-side**				**NYCU parking lot**			
	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
Precision	YOLO-ORE	94.51%	90.16%	97.41%	98.85%	89.50%	79.32%	96.21%	98.86%
	YOLO-SVMv2	89.86%	81.52%	96.53%	98.50%	84.19%	70.76%	92.35%	98.58%
	Increment	**4.65%**	**8.64%**	**0.88%**	**0.35%**	**5.31%**	**8.56%**	**3.86%**	**0.28%**
Recall	YOLO-ORE	89.37%	88.30%	74.34%	98.20%	93.85%	95.87%	85.97%	96.67%
	YOLO-SVMv2	90.68%	88.90%	80.12%	98.35%	94.64%	95.87%	87.58%	97.78%
	Increment	-1.31%	-0.60%	-5.78%	-0.15%	-0.79%	**0.00%**	-1.61%	-1.11%
Accuracy	YOLO-ORE	84.96%	80.54%	72.90%	97.09%	84.54%	76.59%	83.15%	95.61%
	YOLO-SVMv2	82.26%	74.04%	77.87%	96.89%	80.37%	68.66%	81.65%	96.42%
	Increment	**2.70%**	**6.50%**	-4.97%	**0.20%**	**4.17%**	**7.93%**	**1.50%**	-0.81%

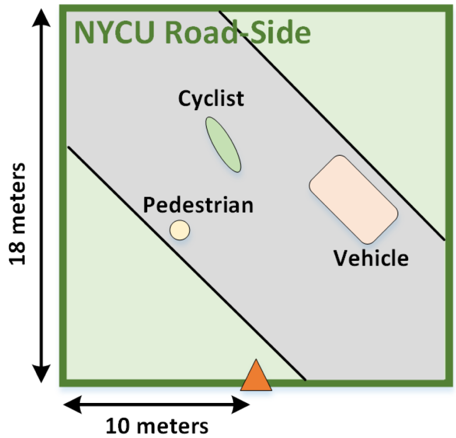


Fig. 17. Illustration of the NYCU road-side scenario.

more accurate classification for a specific class, e.g., pedestrian, the system would trust the SVM result whenever SVM classifies the object to that specific class. Following this concept, we conduct modification of the SVM part in [37] such that it can be used for object detection of more than two objects by first empirically checking whether SVM can provide more accurate classification for a specific class of objects, and then trust the SVM results each time SVM classifies the object to that specific class. The object detection approach after our modification is then called YOLO-SVMv2 below. We note that because YOLO-SVMv2 is, to the best of our knowledge, the best learning-aided object recognition approach for radar systems, it is used as the main comparison approach in this paper. The comparison results with another learning-aided approach, i.e., Faster-RCNN [28], are provided in Appendix A. We can see that the proposed YOLO-ORE system significantly outperforms Faster-RCNN in all measurement scenarios.

The results of the proposed YOLO-ORE system and YOLO-SVMv2 system are shown in Tables V and VI, respectively. For the SC scenarios, the accuracy of the proposed YOLO-ORE for SCSR, SCLR, and SCMR are 93.02%, 83.21%, and 83.93%, respectively, and the performance of the proposed YOLO-ORE system is better than the YOLO-SVMv2 system. Similarly, when considering the more complicated MC scenarios, the accuracy of the proposed YOLO-ORE for MCSR, MCLR, and MCMR are 80.35%, 87.02%, and 84.75%, respectively, and the performance of the proposed YOLO-ORE system is again better than the YOLO-SVMv2 system. The overall comparison between the YOLO-ORE and YOLO-SVMv2 systems in terms of precision, recall, and accuracy is shown in Table VII. We see that our experiments validate that both the proposed YOLO-ORE and the YOLO-SVMv2 can provide good object detection performance not only in simple scenarios but also in realistic and complicated scenarios. In addition, the results show that the YOLO-ORE system can outperform the YOLO-SVMv2 system in all scenarios, showing the superiority of our proposed approach. Note that as the computational complexity of both YOLO-ORE and YOLO-SVMv2 mainly comes from computing the embedded YOLO model, their computational complexity is similar. Specifically, when running on a python platform with GPU RTX 2080 Ti and CPU i7-9700, the average runtime per radar image for YOLO-ORE and YOLO-SVMv2 are 0.0477 seconds and 0.0475 seconds, respectively. Finally, although the recall of our proposed YOLO-ORE could be slightly lower than YOLO-SVMv2, the precision is much higher than that of the YOLO-SVMv2. This is because our proposed ORE module can first eliminate the overlaps and objects that should not exist, and then re-classify the objects. Consequently, although some originally correctly classified objects could be accidentally eliminated, leading to the slight reduction of recall, the precision can be significantly improved. This indicates that the proposed YOLO-ORE system can provide a good tradeoff between recall and precision. Furthermore, the proposed YOLO-ORE system indeed can adjust the tradeoff between recall and precision by adjusting the threshold set in Table IV. This thus gives YOLO-ORE system the flexibility to adapt to different considerations and environments.

Based on our experiments, we have several observations. First, when considering the simplest scenario, e.g., the SCSR, the YOLO-ORE system can achieve a very good accuracy of 93.02%. On the other hand, for the most complicated scenario, namely, the MCMR, we can still achieve a good accuracy of 84.75%, though the performance degrades as compared to SCSR due to the complication of the scenario. Second, we see that the MCSR has slightly worse performance than that of the MCMR, even though the MCSR considers only objects with short range. This might be because the short-range objects could exhibit stronger features, and thus the trained machine would be more easily overfit to the training data as compared to that of the MCSR case. Third, we see that in the SCSR scenario, the radar cross-section (RCS) of vehicles at certain angles is much more significant than others. As a result, it leads to that the patterns of vehicles on the RA map become less uniform than those in normal cases, making vehicles harder to detect than pedestrians and cyclists. Also, in MC scenarios, although the above issues of vehicles might be relieved, the detection could suffer from clutters due to reflections from unwanted objects in the environment. We note that the clutter issue is mostly caused by the unwanted reflections of vehicles which in general results in that the radar incorrectly detects the clutters as pedestrians or cyclists. Therefore, the accuracy of pedestrians and cyclists is lower than that of the vehicles when considering the MC scenarios. Finally, we note that although we assume that the overlappings among objects are incorrect in this paper and such assumption is valid for most cases, we indeed see that there are some overlappings, as illustrated in Fig. 13, which are correct in the MCLR scenario; these overlappings are very difficult to be recognized and recovered.

At the end, we evaluate the efficacy of our proposed NMSDC algorithm and FPS modules in terms of the overlap elimination ratio and re-classification accuracy, respectively, where results are shown in Tables VIII and IX, respectively. We see that our proposed NMSDC algorithm can effectively eliminate overlappings. Also, we see that the proposed FPS classifier can provide very high re-classification accuracy for those objects that need re-classifications.

TABLE XI
COMPARISONS BETWEEN THE YOLO-ORE AND THE FASTER-RCNN IN STANDARD SCENARIOS

Dataset		**SCSR**				**SCLR**				**SCMR**			
	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
Pre.	Y.-O.	98.65%	97.52%	99.51%	99.33%	86.96%	91.48%	92.89%	76.55%	91.77%	91.21%	92.81%	91.18%
	F.-RCNN	93.48%	98.90%	88.29%	94.41%	51.46%	83.69%	44.26%	46.81%	85.64%	83.27%	85.84%	87.10%
	Incre.	**5.17%**	-1.38%	**11.22%**	**4.92%**	**35.50%**	**7.79%**	**48.63%**	**29.74%**	**6.13%**	**7.94%**	**6.97%**	**4.08%**
Rec.	Y.-O.	94.21%	99.16%	98.06%	83.15%	95.40%	97.61%	93.85%	94.54%	90.76%	94.89%	94.39%	83.37%
	F.-RCNN	87.62%	75.63%	95.15%	94.94%	77.85%	56.46%	94.87%	84.15%	73.64%	62.90%	69.51%	87.53%
	Incre.	**6.59%**	**23.53%**	**2.91%**	-11.79%	**17.55%**	**41.15%**	-1.02%	**10.39%**	**17.12%**	**31.99%**	**24.88%**	-4.16%
Acc.	Y.-O.	93.02%	96.72%	97.58%	82.68%	83.21%	89.47%	87.56%	73.31%	83.93%	86.95%	87.95%	77.15%
	F.-RCNN	82.58%	75.00%	84.48%	89.89%	44.89%	50.86%	43.22%	43.02%	65.55%	55.85%	62.36%	77.49%
	Incre.	**10.44%**	**21.72%**	**13.10%**	-7.21%	**38.32%**	**38.61%**	**44.34%**	**30.29%**	**18.38%**	**31.10%**	**25.59%**	-0.34%

Dataset		**MCSR**				**MCLR**				**MCMR**			
	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
Pre.	Y.-O.	89.99%	88.16%	92.92%	89.29%	89.59%	85.80%	89.14%	91.67%	90.58%	83.33%	93.15%	94.16%
	F.-RCNN	63.39%	72.34%	56.25%	64.87%	26.29%	24.28%	20.63%	29.88%	83.84%	82.87%	88.19%	82.26%
	Incre.	**26.60%**	**15.82%**	**36.67%**	**24.42%**	**63.30%**	**61.52%**	**68.51%**	**61.79%**	**6.74%**	**0.46%**	**4.96%**	**11.90%**
Rec.	Y.-O.	88.24%	89.63%	81.74%	93.36%	96.81%	92.07%	95.12%	100%	92.94%	89.59%	88.86%	98.14%
	F.-RCNN	83.82%	70.54%	85.89%	95.02%	74.16%	35.98%	71.95%	94.24%	56.74%	36.32%	50.61%	75.30%
	Incre.	**4.42%**	**19.09%**	-4.15%	-1.66%	**22.65%**	**56.09%**	**23.17%**	**5.76%**	**36.20%**	**53.27%**	**38.25%**	**22.84%**
Acc.	Y.-O.	80.35%	80.00%	76.95%	83.96%	87.02%	79.89%	85.25%	91.67%	84.75%	75.98%	83.41%	92.50%
	F.-RCNN	56.48%	55.56%	51.49%	62.74%	24.09%	16.95%	19.09%	29.34%	51.15%	33.78%	47.39%	64.77%
	Incre.	**23.87%**	**24.44%**	**25.46%**	**21.22%**	**62.93%**	**62.94%**	**66.16%**	**62.33%**	**33.60%**	**42.20%**	**36.02%**	**27.73%**

TABLE XII
COMPARISONS BETWEEN THE YOLO-ORE AND THE FASTER-RCNN IN EXTENDED SCENARIOS

Dataset		**NYCU road-side**				**NYCU parking lot**			
	Class	Total	Ped.	Cyc.	Veh.	Total	Ped.	Cyc.	Veh.
Precision	YOLO-ORE	94.51%	90.16%	97.41%	98.85%	89.50%	79.32%	96.21%	98.86%
	Faster-RCNN	81.75%	84.75%	64.57%	87.67%	82.56%	79.29%	72.17%	95.98%
	Increment	**12.76%**	**5.41%**	**32.84%**	**11.18%**	**6.94%**	**0.03%**	**24.04%**	**2.88%**
Recall	YOLO-ORE	89.37%	88.30%	74.34%	98.20%	93.85%	95.87%	85.97%	96.67%
	Faster-RCNN	92.62%	88.97%	88.76%	99.25%	90.19%	93.85%	89.52%	86.78%
	Increment	-3.25%	-0.67%	-14.42%	-1.05%	**3.66%**	**2.02%**	-3.55%	**9.89%**
Accuracy	YOLO-ORE	84.96%	80.54%	72.90%	97.09%	84.54%	76.69%	83.15%	96.61%
	Faster-RCNN	76.30%	76.69%	59.69%	87.09%	75.76%	75.37%	66.55%	83.74%
	Increment	**8.66%**	**3.85%**	**13.21%**	**10.00%**	**8.78%**	**1.32%**	**16.60%**	**12.87%**

D. Experiment Results for Extended Scenarios

In this subsection, we evaluate our proposed YOLO-ORE system and compare it to the YOLO-SVMv2 system in two extended scenarios: NYCU parking lot scenario and the NYCU road-side scenario. Exemplary radar images and their corresponding camera images for the NYCU parking lot and the NYCU road-side scenarios are shown in Figs. 14 and 15, respectively. Besides, the illustrations of the NYCU parking lot and NYCU road-side scenarios are provided in Figs. 16 and 17, respectively. The reasons for providing these two extended datasets are two-fold: (i) to provide performance validation in more diverse scenarios; and (ii) to validate that the YOLO-ORE system can outperform the YOLO-SVMv2 system even in the measurement scenario, i.e., the NYCU parking lot scenario, similar to that of [37].

The radar configuration used for measurements in NYCU parking lot and NYCU road-side scenarios follows the configuration of SR, with the exception that the frequency slope is changed to 40 MHz/us, resulting in 27.63 meters of the maximum range. We again consider three different types of objects, namely, the pedestrian, cyclist, and vehicle, for the measurements. For the NYCU parking lot scenario, we in total collect 2040 radar images, and then randomly choose 1314 radar images along with the MCSR dataset to train the systems. The remaining 726 radar images are used for testing. Note that the training with the aid of the MCSR dataset is to guarantee that the training can converge without overfitting. For the NYCU road-side scenario, we in total collect 1818 radar images, and then randomly choose 1206 radar images along with the MCSR dataset to train the systems. The remaining 612 radar images are then used for testing. We note that the NYCU road-side dataset is to validate our system in practical road-side scenarios.

The results of both NYCU parking lot and NYCU road-side datasets are shown in Table X. From the results, we again see that the proposed YOLO-ORE system has better accuracy than that of the YOLO-SVMv2 system in both NYCU parking lot and NYCU road-side scenarios.

V. CONCLUSION

In this paper, we proposed a novel deep learning-aided object recognition approach, called YOLO-ORE, by combining YOLO with the proposed radar image generation module and object recheck system. Our proposed ORE system was used to deal with the drawbacks of YOLO in which the overlap errors between different classes of objects and the misclassification errors of the low-confidence objects could happen. We conducted extensive real-world experiments in realistic scenarios to evaluate our proposed approach. Results showed that our proposed YOLO-ORE system outperforms the state-of-the-art YOLO-based object recognition system. In addition, the results validate that the proposed learning-aided object recognition approach can work well in realistic scenarios.

Appendix A
Comparison Results Between YOLO-ORE and Faster-RCNN

In addition to the YOLO-SVM system proposed in [37], to the best of our knowledge, the best machine learning-aided radar object recognition approaches are the Faster-RCNN and SSD mentioned in [28]. Then, because according to the investigation in [28], the performance of Faster-RCNN is better than SSD, we compare the Faster-RCNN with our YOLO-ORE system. The results are shown in Tables XI and XII. Results show that our YOLO-ORE system significantly outperforms the Faster-RCNN in all scenarios.

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